🏥 Hospital Length of Stay (LOS) Predictor

Enterprise demo: XGBoost regression • cost-safe MLOps workflow • SHAP explainability • local UI

Samples	Features	Val MAE (days)	Val RMSE (days)
101,766	130	1.610	2.229

🔮 Predict 🧠 Explain (SHAP) 📌 Project Notes

Explainability (SHAP)

These assets are generated by `python src/explainability/shap_summary.py` and saved under `reports/`.

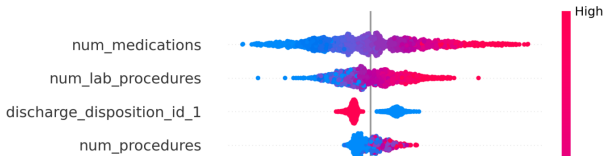
Top Drivers of LOS (Global Explainability Summary)

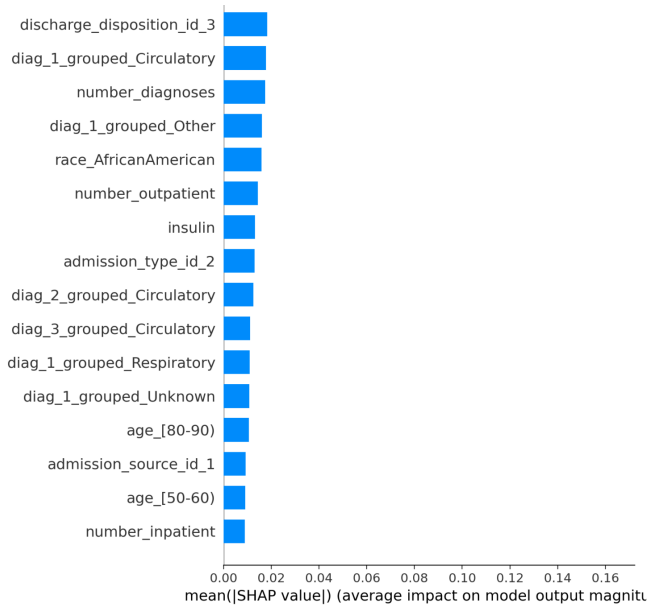
Ranked by **mean absolute SHAP value** (higher = more impact on the model's LOS prediction). This is a standard enterprise explainability summary.

	rank	feature
0	1	num_medications
1	2	num_lab_procedures
2	3	discharge_disposition_id_1
3	4	num_procedures
4	5	discharge_disposition_id_3
5	6	diag_1_grouped_Circulatory
6	7	number_diagnoses
7	8	diag_1_grouped_Other
8	9	race_AfricanAmerican
9	10	number_outpatient

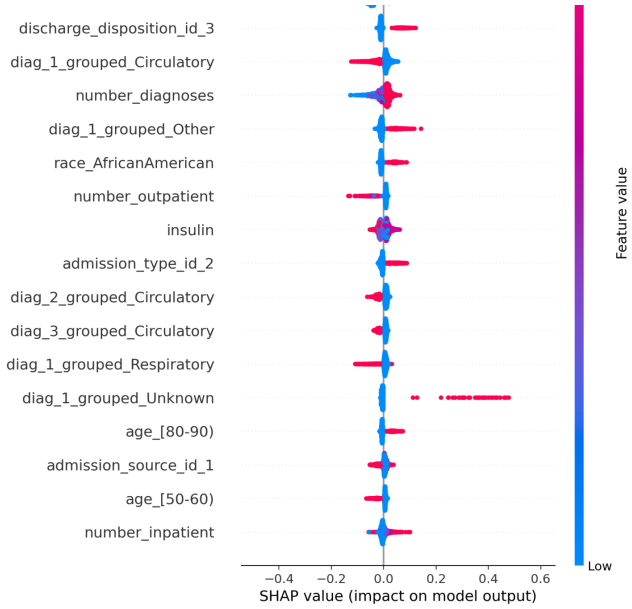
Download top feature table (CSV)

Global SHAP Plots





Global Importance (Bar)

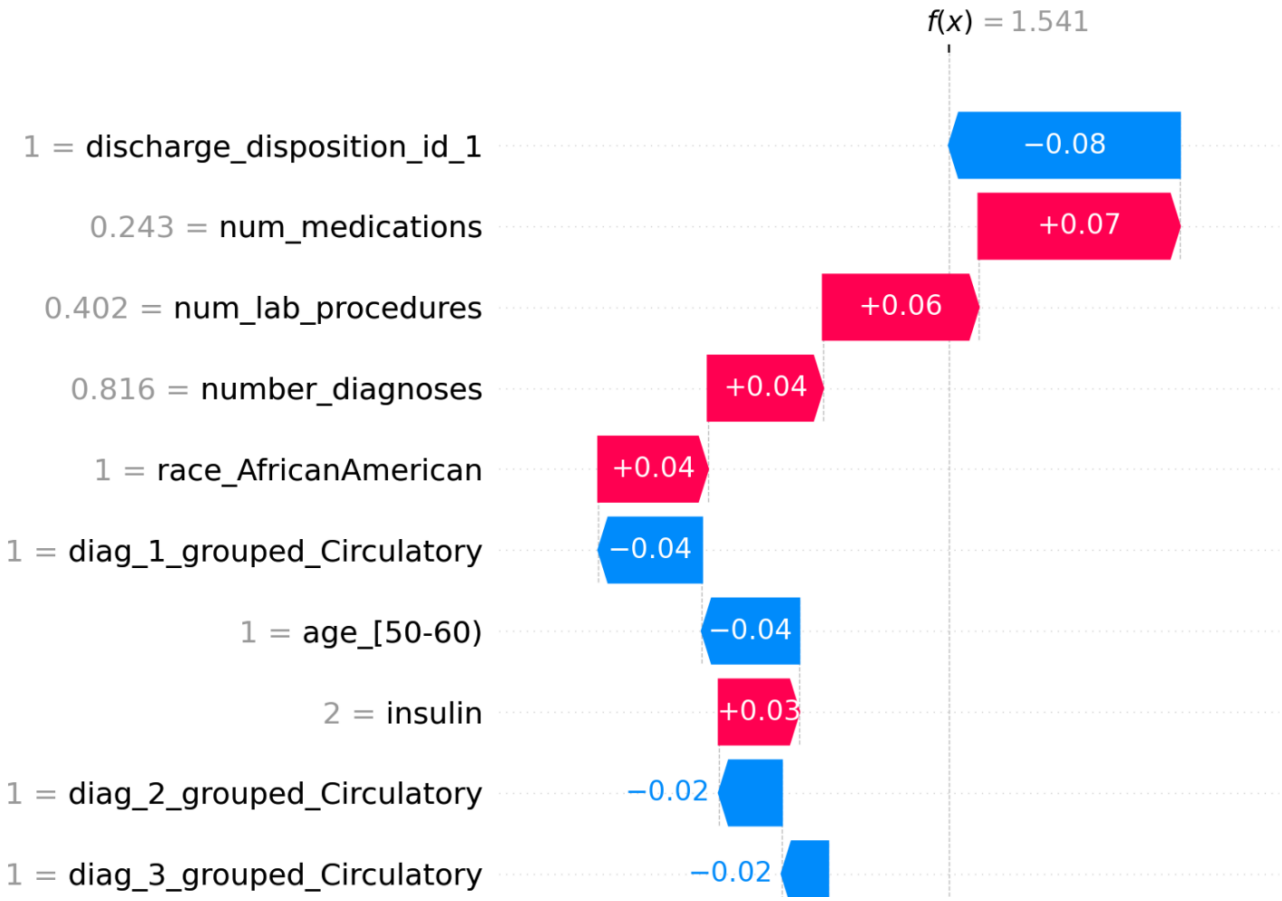


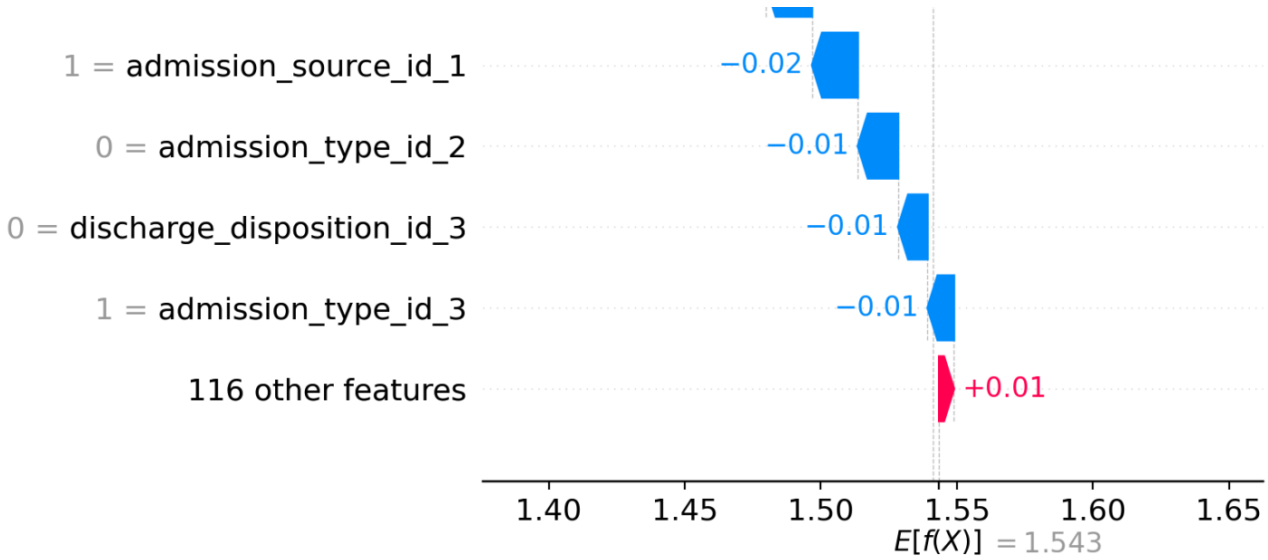
Impact Distribution (Beeswarm)

Download bar plot

Download beeswarm plot

Per-Patient Explanation (Local)



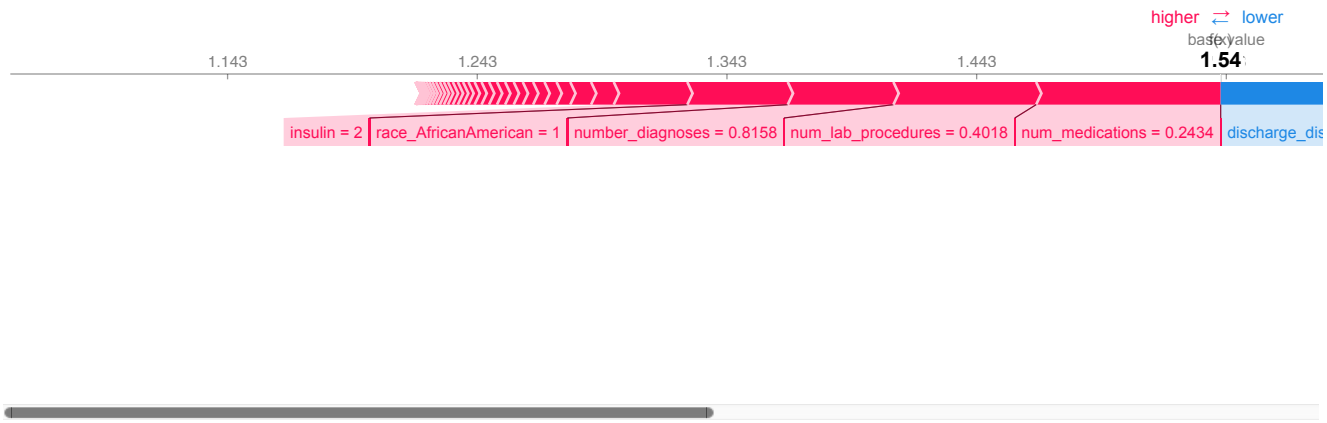


Waterfall Explanation (Single Example)

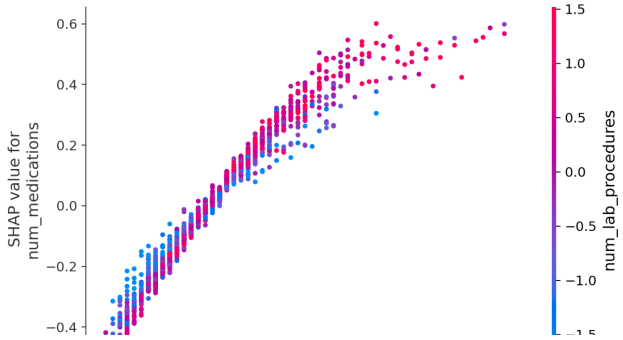
Download waterfall plot

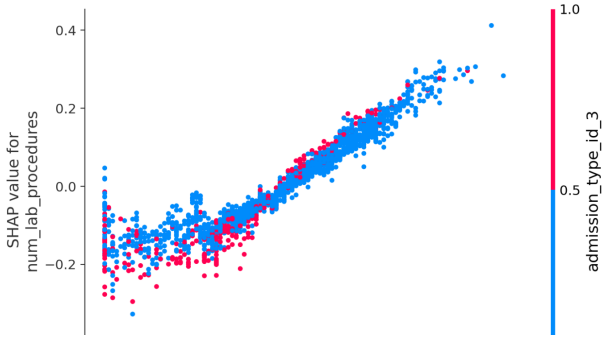
Force Plot (Interactive)

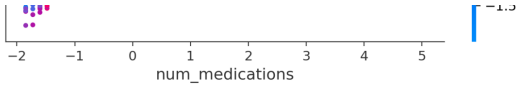
Download force plot HTML



Feature Behavior (Dependence Plots)

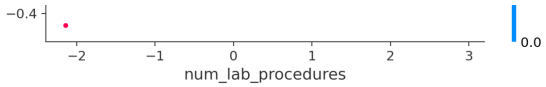






Dependence Plot (Top feature #1)

Download dependence plot #1



Dependence Plot (Top feature #2)

Download dependence plot #2